

Baseline Information

Transit Rate Tables - Per Site (Saudi Market + Freightos estimates)

Batch Size 1000
 Unit weight (kg/wafer): \$ 0.32 According to resource
 Batch weight (kg per 100,000): 320
 Cost per 1,000 wafers = Rate(B5) x Batch Weight(B6). Batch = 1,000 wafers = 40 FOUPs = 320 kg (8 kg/load)

Baseline Assumptions

Country Premium

Country	USA	JPN
Rates	1.30	1.2

Cost Assumptions Per Wafer

	A Cost		B Cost		C Cost
\$	20,000.00	\$	16,000.00	\$	9,000.00

Demand Forecast

DEMAND (300mm wafers / year)
 Global total wafers (TSMC Website): 17000000 ABC Total
 A=below3nm, B=5nm, C=7nm 0.25

Demand Forecast Allocation

Per City Allocation (W/Ratio)			33.78%
City	Share	Product A	
Silicon Valley		0.1	425000
Austin		0.1	425000
Shenzhen		0.3	1275000
Seoul		0.35	1487500
Munich		0.1	425000
Tokyo		0.05	212500
TOTAL		1	4250000

N

HSINCHU (Taiwan) - makes A, B, C

Shipping

Destination	Rate (\$/kg)	Cost per 1,000
Silicon Valley	\$ 6.48	\$ 2,073.60
Austin	\$ 7.52	\$ 2,406.40
Shenzhen	\$ 2.48	\$ 793.60
Seoul	\$ 3.52	\$ 1,126.40
Munich	\$ 6.00	\$ 1,920.00
Tokyo	\$ 3.52	\$ 1,126.40

TAINAN (Taiwan) - makes A, B, C

Destination	Rate (\$/kg)	Cost per 1,000
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Silicon Valley	\$	6.48	\$	2,073.60
Austin	\$	7.52	\$	2,406.40
Shenzhen	\$	2.48	\$	793.60
Seoul	\$	3.52	\$	1,126.40
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	3.52	\$	1,126.40

KAOHSIUNG (Taiwan) - makes A

Destination	Rate (\$/kg)		Cost per 1,000	
Silicon Valley	\$	6.48	\$	2,073.60
Austin	\$	7.52	\$	2,406.40
Shenzhen	\$	2.48	\$	793.60
Seoul	\$	3.52	\$	1,126.40
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	3.52	\$	1,126.40

ARIZONA (US) - makes A, B

Destination	Rate (\$/kg)		Cost per 1,000	
Silicon Valley	\$	2.48	\$	793.60
Austin	\$	2.00	\$	640.00
Shenzhen	\$	7.04	\$	2,252.80
Seoul	\$	6.48	\$	2,073.60
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	6.00	\$	1,920.00

Taichung - makes B,C

Destination	Rate (\$/kg)		Cost per 1,000	
Silicon Valley	\$	6.48	\$	2,073.60
Austin	\$	7.52	\$	2,406.40
Shenzhen	\$	2.48	\$	793.60
Seoul	\$	3.52	\$	1,126.40
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	3.52	\$	1,126.40

JAPAN - makes A, B, C- Note that this cannot be made right now but it is important to know that this is a good

Destination	Rate (\$/kg)		Cost per 1,000	
Silicon Valley	\$	6.48	\$	2,073.60
Austin	\$	7.52	\$	2,406.40
Shenzhen	\$	3.04	\$	972.80
Seoul	\$	2.48	\$	793.60
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	1.52	\$	486.40

Links For Info and Resources

Revenue-> Demand

17 Million wafers

Wafer Price

Air Freight

Air Freight

<https://investor.tsmc.com/english/encrypt/file>

<https://www.tsmc.com/english/dedicatedFour>

<https://siliconanalysts.com/guide/semiconduc>

<https://suaidglobal.com/insights/air-freight-co>

<https://www.freightos.com/freight-resources/f>

s

led 25-wafer FOUP, per Wikipedia/Entegris/Semitracks). Est., not TSMC contract rates.- This is our Assumptions*

Taiwan Semi- ABC~74.5(Multiply by Ratio) 1.351351351

12580000

0.36 0.13 ~Note this is 74% of prduct mix

	48.65%	17.57%
Product B	Product C	
	612000	221000
	612000	221000
	1836000	663000
	2142000	773500
	612000	221000
	306000	110500
	6120000	2210000
	12580000	Total Demand

ode Application

Product Cost to Make				Total Costs	
	A	B	C	Total Variable A	Total Variable B
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,002,073.60	\$ 16,002,073.60
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,002,406.40	\$ 16,002,406.40
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,000,793.60	\$ 16,000,793.60
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,001,126.40	\$ 16,001,126.40
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,001,920.00	\$ 16,001,920.00
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,001,126.40	\$ 16,001,126.40

\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,002,073.60	\$	16,002,073.60
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,002,406.40	\$	16,002,406.40
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,000,793.60	\$	16,000,793.60
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,001,126.40	\$	16,001,126.40
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,001,920.00	\$	16,001,920.00
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,001,126.40	\$	16,001,126.40

\$	20,000,000.00		\$	20,002,073.60
\$	20,000,000.00		\$	20,002,406.40
\$	20,000,000.00		\$	20,000,793.60
\$	20,000,000.00		\$	20,001,126.40
\$	20,000,000.00		\$	20,001,920.00
\$	20,000,000.00		\$	20,001,126.40

\$	26,000,000.00	\$ 20,800,000.00		\$	26,000,793.60	\$	20,800,793.60
\$	26,000,000.00	\$ 20,800,000.00		\$	26,000,640.00	\$	20,800,640.00
\$	26,000,000.00	\$ 20,800,000.00		\$	26,002,252.80	\$	20,802,252.80
\$	26,000,000.00	\$ 20,800,000.00		\$	26,002,073.60	\$	20,802,073.60
\$	26,000,000.00	\$ 20,800,000.00		\$	26,001,920.00	\$	20,801,920.00
\$	26,000,000.00	\$ 20,800,000.00		\$	26,001,920.00	\$	20,801,920.00

	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,002,073.60
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,002,406.40
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,000,793.60
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,001,126.40
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,001,920.00
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,001,126.40

1 alternative for the future

\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,002,073.60	\$	19,202,073.60
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,002,406.40	\$	19,202,406.40
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,000,972.80	\$	19,200,972.80
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,000,793.60	\$	19,200,793.60
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,001,920.00	\$	19,201,920.00
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,000,486.40	\$	19,200,486.40

Transport Matrix	Product(2/3 NM)			
<i>Fab Location(Cities)</i>	Silicon	Aus	Shz	Seo
Hsinchu	2073.6	2406.4	793.6	1126.4
Tainan	2073.6	2406.4	793.6	1126.4
Kaohsiung	2073.6	2406.4	793.6	1126.4
Arizona	793.6	640	2252.8	2073.6
Taichung				
Japan				
TOTAL Demand	425	425	1275	1487.5

Product Cost Matrix	A Product(2/3 NM)			
<i>Fab Location(Cities)</i>	Silicon	Aus	Shz	Seo
Hsinchu	20000000	20000000	20000000	20000000
Tainan	20000000	20000000	20000000	20000000
Kaohsiung	20000000	20000000	20000000	20000000
Arizona	26000000	26000000	26000000	26000000
Taichung				
Japan				

Total Variable C

\$ 9,002,073.60
 \$ 9,002,406.40
 \$ 9,000,793.60
 \$ 9,001,126.40
 \$ 9,001,920.00
 \$ 9,001,126.40

\$ 9,002,073.60
\$ 9,002,406.40
\$ 9,000,793.60
\$ 9,001,126.40
\$ 9,001,920.00
\$ 9,001,126.40

\$ 9,002,073.60
\$ 9,002,406.40
\$ 9,000,793.60
\$ 9,001,126.40
\$ 9,001,920.00
\$ 9,001,126.40

\$ 10,802,073.60
\$ 10,802,406.40
\$ 10,800,972.80
\$ 10,800,793.60
\$ 10,801,920.00
\$ 10,800,486.40

Node Matrix Application

B Product(5 NM)						C Product(7 NM)			
Mun	Tok	Silicon	Aus	Shz	Seo	Mun	Tok	Silicon	Aus
1920	1126.4	2073.6	2406.4	793.6	1126.4	1920	1126.4	2073.6	2406.4
1920	1126.4	2073.6	2406.4	793.6	1126.4	1920	1126.4	2073.6	2406.4
1920	1126.4								
1920	1920	793.6	640	2252.8	2073.6	1920	1920		
		2073.6	2406.4	793.6	1126.4	1920	1126.4	2073.6	2406.4
425	212.5	612	612	1836	2142	612	306	221	221

[illegible]

				Associated Costs Constraint
Shz	Seo	Mun	Tok	Fixed Value (Billions)
793.6	1126.4	1920	1126.4	12
793.6	1126.4	1920	1126.4	12
				9.49
				65
793.6	1126.4	1920	1126.4	2.87
				24
663	773.5	221	110.5	

C Product(7 NM)			
Shz	Seo	Mun	Tok
9000000	9000000	9000000	9000000
9000000	9000000	9000000	9000000
9000000	9000000	9000000	9000000

Constraint (Millions)
1.2
3
1.44
0.72
1.2
1.2